Daniel Hardesty Lewis

Forward Deployed Engineer – full-stack, enterprise operations, AI & automation

Engineer and founder specializing in bridging product engineering and enterprise stakeholders. Built full-stack platforms, automated operational workflows, and shipped customer-facing tooling from scoping through launch.

Core Expertise

8 years **Full-Stack Engineering**: Ship production web applications end-to-end using Python, React, SQL, and Postgres; own from data pipeline through user-facing delivery

4 years **Client Engineering & Onboarding**: Triage enterprise requests, scope integrations, and own technical onboarding from kickoff through launch

8 years **Process Automation**: Find manual bottlenecks and replace them with automated workflows and AI tooling; build repeatable playbooks for client engagement

8 years **Communication & Education**: Teach customers how to integrate, keep stakeholders current on project status, and document institutional knowledge

8 years **Technical Leadership**: Bridge engineering and stakeholders; mentor engineers through design reviews, code reviews, and hands-on instruction

Technologies

Core stack	Python; SQL; TypeScript; React; Postgres; Redis; Elasticsearch; Supabase; FastAPI
AI/ML	LangChain; LangGraph; DSPy; RAG; FAISS; PyTorch; scikit-learn; Airflow
Data	Pandas; Polars; PyArrow; Parquet; BigQuery; PostGIS; ETL pipelines
Infrastructure	Linux; Git; Docker; Kubernetes; Terraform; Modal; GCS; CI/CD; monitoring & observability
Additional	CUDA; C++; Ray; Dask; Spark

Experience

Professional

Nov. 2024 – Present **Co-founder**, *Homecastr*, New York, NY

Consumer real-estate pricing and forecasting platform with interactive uncertainty fancharts and temporal leakage-safe expanding-window evaluation.

- Shipped a nationwide, tract-level forecasting system, achieving an 8% one-year forecast error that beats Zillow's 8.4% industry benchmark, and proving long-term stability with a 25% median error at a 4-year horizon (a timescale competitors don't offer).
- Built an advanced diffusion model from the ground up, designing custom architecture to ensure realistic price trajectories and reliable confidence intervals across the United States.
- Created a custom evaluation framework to validate model confidence and reliability (coverage, sharpness, and PIT histograms) prior to user delivery.
- Owned the complete technical workflow—from data engineering to the live web UI (homecastr.com)—while refining go-to-market strategy based on feedback from top proptech investors including a16z and MetaProp.

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation; led **customer-facing scoping, demos, and technical onboarding** for enterprise prospects.

- Specified an **agentic workflow** and schema definitions for BOM parsing, retrieval, and tariff simulation.
- Directed a **conversational AI interface** enabling non-technical decision-makers to retrieve BOMs and receive proactive compliance alerts without engineering support.
- Ran **customer discovery and live demos** with Deloitte, ScoutyAI, and other import/export founders; **scoped integration requirements** for enterprise prospects.
- Contributed to materials presented to KPMG, pre-dating their launch of a tariff-modeling product. Ongoing discovery with an ex-Goldman Sachs co-CIO and Columbia Business School tariff economists.

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Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY
The highest quality seamless elevation data for Texas; web delivery for engineering and hazard workflows.
- Engineered a statewide seamless Texas DEM mosaic, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across 70+ state, federal, and local elevation datasets.
- Built the **web distribution platform end-to-end**: data pipeline, tiling, hosting, and delivery UX.
- Commercial discussions to license elevation datasets to research and AI organizations.

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- **Triaged and scoped enterprise requests** across 6+ state and federal agencies including NOAA, NSF, USACE, and GLO, translating requirements into SOW and MOU frameworks.
- Served as a primary **technical point of contact** for the $40M Texas Disaster Information System (TDIS), coordinating between engineering, Texas Emergency Management, and first responders.
- Documented data onboarding so partner agencies could **self-serve integrations** without engineering support.
- Managed $1M+ compute allocations across 10+ projects, orchestrating 800,000-node distributed workloads on leadership-class supercomputers for climate and flood modeling.
- Spun out Summit Geospatial alongside UT's VPs of IP to commercialize the platform's elevation data layer.
- **Evaluated and accepted** $10M+ in computing resource requests as Technical Reviewer for Frontera Doctoral Fellows and Large-Scale Computing Pathways, triaging proposals against capacity and impact criteria.

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX
- Competed in year-long DARPA World Modelers program to improve disaster resiliency and food security modeling in East Africa, coordinating across distributed engineering and research teams.
- Received recognition from AAAI President Yolanda Gil during her farewell address for contributions to automated scientific modeling.

Research

Jan. 2025 – Oct. 2025 **Industry-sponsored research assistant to Ask2.AI CSO Ali Hirsa**, *Columbia University*, New York, NY
- Developed a multi-asset CVAE latent factor model for **tabular financial data** with Skew-T Mixture priors; achieved 85% R^2 vs. commercial SaaS (75%) and Fama-French (58%) on backtested holdout.
- Derived a tractable, axiom-compliant per-feature SHAP algorithm using Hessian-based recursive clustering to provide stable temporal regime-dependent attributions for deep learning models.

May 2025 – Aug. 2025 **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
Math benchmarking and finetuning of Deepseek's GRPO algorithm.

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
Integrated hydrological models into MINT's scientific orchestration platform submission to DARPA.

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras geoscientists and petroleum engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, and HPC.

Select Publications

2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), "Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making". *ACM TiiS*, 2021. DOI: 10.1145/3453172

2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), "An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models". *ACM IUI '19*, 2019. DOI: 10.1145/3308557.3308711

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Key Projects

oppsAlert Developed the first property- and owner-level ML model of NIMBYism, comparing diffusion against logistic regression, using 15+ years of 15,000+ homeowner protest records from the City of Austin.

Activities

Sailing Volunteer Atlantic SAR of the Cheeki Rafiki

Bikepacking US West & East Coasts

Languages

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Spanish Native Proficiency

French Professional Proficiency

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